MIL-S-10133B
2 OCTOBER 1956
SUPERSEDING
MIL-S-10133A
18 JUNE 1952

MILITARY SPECIFICATION

SEAT, OUTLET-VALVE, ALUMINUM-BASE-ALLOY DIECASTING FOR OUTLET VALVE, C15

This specification has been approved by Department of Defense *and is mandatory for use by the Departments of the Army, the Navy and the Air Force.*

1. SCOPE

1.1 This specification covers an outlet valve seat for use on gas masks,

2.. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT-P-1 41 — Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing

LLL-B-631 — Boxes; Fiber, Corrugated (for Domestic Shipment.

STANDARDS

MILITARY

MIL-STD-10 — Surface Roughness, Waviness, and Lay

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

DRAWINGS

CHEMICAL COWS

D5-2-606 — Seat, Outlet Valve.

PUBLICATIONS

CHEMICAL CORPS DIRECTIVE

567 — operation and Maintenance of Outlet Valve Leakage In) dicator, IE14.

(Copies of specifications, standards, drawings, and pubications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. All materials used in the manufacture of the seat shall conform to the specifications and drawings forming a part of this specification.

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3.2 Manufacture. The seat shall be manufactured as shown on Drawing D5-2-606 showing a detail of Drawing E5-2-597. Surface roughness specified by symbols on detail drawing shall comply with the applicable classes in Standard 31 MIL-STD-10.

3.3 Physical requirements.

3.3.1 Rigidity. The hub (see fig. 1) shall depress not more than 0.016 inch when tested as specified in 4.4.1.

3.3.2 *Impact resistance.* The seat shall not break nor shall the ribs or hub be deformed so that any portion of the flat surface of the hub is below the upper edge of the sealing rim surface when tested as specified in 4.4.2.

3.3.3 *Leakage*. The leakage through the seat, before and after corrosion testing, shall be no more than 15 milliliters (ml.) of air per minute when tested as specified in 4.4.3.

3.3.4 *Corrosion resistance.* The seat shall show no corrosion when tested as specified in 4.4.4.

3.4 Workmanship. The portions of the seat designed to come in contact with rubber parts shall be free of sharp edges, The seat shall be substantial}}- free from flash.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of not more than the production of each die of each machine for a period not exceeding 48 hours continuous production. Any shutdown or any alteration in the continuity of production, such as a change of machine, die, modification of die, or change of pot or composition of alloy, shall be considered as the start of a new lot, Diecastings shall be kept separated by lot for trimming, finishing, and shipping. The inspection lot for nondestructive inspection, within the above limitations, shall be designted by the inspector.

4.2 Sampling.

4.2.1 For nondestructive inspection. Sampling for nondestructive inspection shall be conducted in accordance with the authorized classification of defects (1.3.3) and Standard MIL-STD-105.

4.2.2 For destructive tests,

4.2.2.1 *lmpact* resistance *and rigidity* Sixty-six seats, 22 for test and 44 for retest, shall be taken at random from each lot for tests as specified in 4.4.1 and 4.4.2.

4.2.2.2 *Corrosion resistance.* Twenty-five seats which have passed the leakage test shall be taken for the corrosion resistance test as specified in 4.4.4.

4.3 Inspection.

4.3.1 *Molds.* The inspector shall ascertain that each mold has been approved by Inspection Equipment Agency, Army Chemical Center, Md., prior to starting production.

4.3.2 *Packaging, packing, and marking.* The inspector shall inspect the packaging, packing, and marking for compliance with section 5.

4.3.3 *Classification of defects.* Inspection shall be conducted as follows:

4.3.3.1 *Seat, outlet valve. (Dwg.* D5-2-606 which is listed on Dwg. E5-2-597.)

Categories and defects	*Acceptance standards*
	AQL 0.25 percent defective
Major A	
1a. Leakage	4.4.3
	AQL 0.65 percent defective
Major	
1. Non-fills	
2. Axial thickness of *hull* at hole incorrect	
3. Height of hub in relatlon to sealing rim surface incorrect	
4. Hub *hole* diameter incorrect	
5. Senling rim surface rough, irregular, burred, or cracked	

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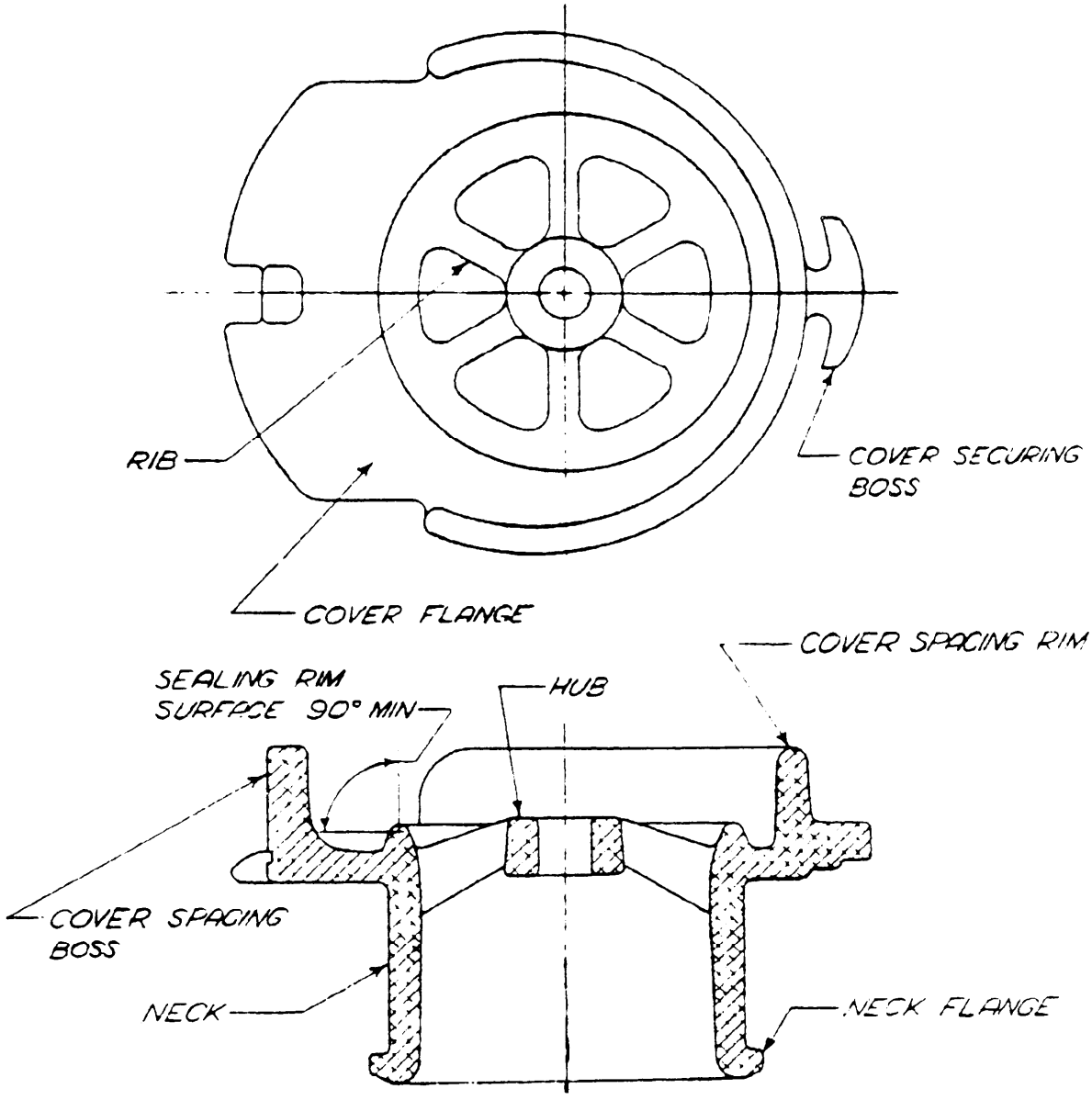


FIGURE 1. Outlet valve seat

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Categories and defects (continued):

AQL 0.65 percent defect

Major (continued)

6. Neck flange sharp edge
7. Neck cracked
8. Contamination (greases and oil)
9. Hub cracked, burrs, breaks, rough or sharp edges
10. Chamfer excessive at lower hub surface

AQL 1.0 percent defectivc

Minor

51. Overall height Incorrect
52. Cracks, burrs, breaks, rough or sharp edges
(a) Cover spacing rim
(b) Cover flange
(c) *Cover* spacing boss
(d) Cover securing boss
(e) Neck flange (other than **Major** defect No. 6)
(f) Ribs
53. Excessive mold flash
54. Nech outs.ide diameter incorrect
55. Coating missing or inadequate

AQL 2.5 percent defective

Incidental

101. Contamination (other than Major defect No. 8)
104. Color incorrect
103. Identification marking missing or illegible

4.4 Tests

4.4.1 Rigidity Place the seat on the jig in the position shown in figure 2. Slowly apply a load of 50 pounds to the flat surface of the hub. Remove the weight after 1 minute and measure any depression with a micrometer.

4.4.2 *Impact resistance.* Place the seat on the jig in the position shown in figure 2. Drop a 1/4-pound cylindrical steel pellet, 1 inch in diameter with a spherical end, from a height of 24 inches onto the top of the hub. Observe the seat for compliance with 3.3.2.

4.4.3 Leakage. A dry disc (Dwg. E5-2-600) shall be assembled to a dry seat as shown on Drawing E5-2-597 and the assembly tested for leakage in accordance with Directive No. 567.

4.4.4 *Currosion resistance* Determine corrosion resistance in accordance with method 606.1 of Specification TT-P-141. The test shall be run on the seats (after leakage tests) and the duration of the test shall be 120 consecutive hours, After examining for corrosion the seats shall be washed with tap water, dried and retested for leakage (4.4.3).

4.5 **Acceptance criteria.**

4.5.1 *Impact* ***resistance and rigidity.*** If one or two seats fail the tests specified in 4.4.1 and 4.4.2 repeat the tests using the specimens reserved for retest (4.2.2.1). Reject the lot if more than two failures are observed **in** test plus retest.

4.5.2 *Leakage and* corrosion *resistance.*. If any specimen fails the leakage or corrosion tests reject the lot represented.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level B.* one gross of seats shall be packaged in a style RSC double-faced corrugated fiberboard box conforming to Specification LLL-B-631, The inside dimensions of the box shall be 12 inches long by 22 inches wide by $63/8_{8}$ inches deep. The box shall be fitted with interlocking fiberboard partitions and the layers shall be separated by fiberboard pads. Closure shall be by adhesive, tape, or metal fasteners.

5.1.2 Level C. Packaging shall be as specified in 5.1.1.

5.2 Packing .

5.2.1 ***Level B.*** A uniform number of packages (5.1.1) shall be packed in a corrugated fiberboard box conforming to style RSC of Specification LLL-B-631. Closure shall be by adhesive, tape, or metal fasteners,

5.2.2 Level C. Pack shall be as specified in 5.2.1.

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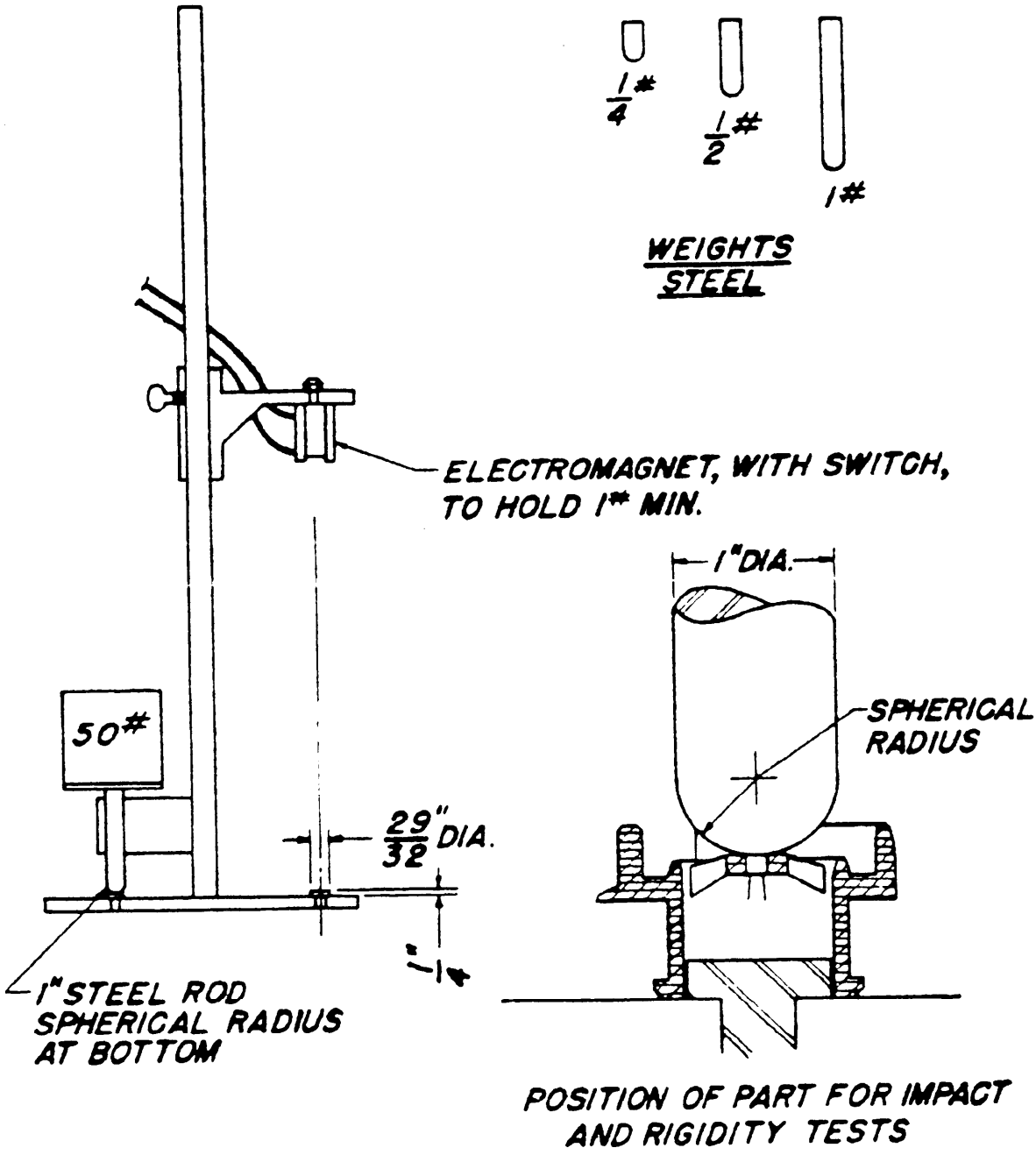


FIGURE 2. *Impact and rigidity testing apparatus*

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5.3 Marking. In addition to any special marking required by contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129

6. 2 NOTES

6.1 Ordering data. Procurement documents should specify—

(a) The title, number, and date of this specification and drawings.

(b) The level of packaging and packing required (see 5.1 and 5.2).

6.2 Sampling and testing. When the contractor consistently produces high quality material and operates under a system of quality control acceptable to the Government, the Government at its discretion, may modify in whole or in part, the sampling and testing procedures sypecified herein However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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